Lesson 5: If — Yes or No? 🔀

> **Goal:** Use `if` and `else` to make decisions in your programs based on conditions.

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1. Making Decisions

Every day you make decisions: "If it is raining, take an umbrella. Otherwise, leave it at home."

Programs make decisions too! In Scratch, you used the **"if/else"** block. In Python, we use `if` and `else`.

Scratch	Python
`if <condition>` block	`if condition:`
`else` block	`else:`
Diamond-shaped condition	Any expression that is True or False

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2. Comparison Operators

Before we can make decisions, we need to **compare** values. Comparison operators return `True` or `False`.

Operator	Meaning	Example	Result
`==`	equal to	`5 == 5`	`True`
`!=`	not equal to	`5 != 3`	`True`
`<`	less than	`3 < 5`	`True`
`>`	greater than	`5 > 3`	`True`
`<=`	less than or equal	`5 <= 5`	`True`
`>=`	greater than or equal	`6 >= 5`	`True`

```
print(5 == 5)     # True
print(5 == 6)     # False
print(3 < 10)     # True
print(10 < 3)     # False
print(7 != 7)     # False
```

> ⚠️ **Important:** `==` (two equals signs) means "is equal to". `=` (one equals sign) means "assign a value". They are different!

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3. Boolean Values: True and False

The result of a comparison is a **boolean** — either `True` or `False`. These are special values in Python (with capital first letters).

```
x = True
y = False
print(x)          # True
print(type(x))    # <class 'bool'>

result = 10 > 5
print(result)     # True
```

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4. The `if` Statement

The `if` statement runs a block of code **only if** the condition is `True`.

```
age = 15

if age >= 18:
    print("You can vote!")
```

Output: (nothing — because 15 is not ≥ 18)

```
age = 20

if age >= 18:
    print("You can vote!")
```

Output:

```
You can vote!
```

The structure:

```
if condition:
    code to run if True
```

```
    more code if True
```

> 🚨 **CRITICAL: Indentation!** The code inside `if` must be indented — use **4 spaces**. Python uses indentation to know which lines belong to the `if` block. This is different from Scratch, where blocks are visually nested.

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5. The `if` / `else` Statement

`else` runs when the condition is `False`.

```
age = 15

if age >= 18:
    print("You can vote!")
else:
    print("You cannot vote yet.")
```

Output:

```
You cannot vote yet.
```

The structure:

```
if condition:
    code to run if True
else:
    code to run if False
```

Only **one** branch runs — either the `if` block or the `else` block, never both.

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6. Indentation is Critical

```
# CORRECT — 4 spaces of indentation
score = 85
if score >= 60:
    print("You passed!")    # 4 spaces before print
else:
    print("Try again.")     # 4 spaces before print

# WRONG — no indentation (causes IndentationError)
```

```
if score >= 60:
print("You passed!")        # ❌ Error!
```

> 💡 In Thonny, press **Tab** to indent automatically. Thonny will add 4 spaces for you.

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7. Common Patterns

Check even or odd:

```
number = int(input("Enter a number: "))

if number % 2 == 0:
    print("Even")
else:
    print("Odd")
```

Compare two numbers:

```
a = int(input("Enter a: "))
b = int(input("Enter b: "))

if a > b:
    print("a is larger")
else:
    print("b is larger or equal")
```

Pass or fail:

```
score = int(input("Enter your score: "))

if score >= 60:
    print("Pass")
else:
    print("Fail")
```

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8. `if` Without `else`

You don't always need an `else`. Sometimes you only want to do something if a condition is True, and do nothing otherwise.

```
temperature = int(input("Temperature: "))

if temperature > 35:
    print("Warning: very hot today!")

print("Have a nice day!")   # This always runs
```

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9. Common Mistakes

```
# Mistake 1: Using = instead of ==
if x = 5:       # ❌ SyntaxError! Use == for comparison
    print("yes")

# Mistake 2: Forgetting the colon :
if x > 5        # ❌ SyntaxError! Need : at the end
    print("yes")

# Mistake 3: Wrong indentation
if x > 5:
print("yes")    # ❌ IndentationError!

# Correct:
if x > 5:
    print("yes")   # ✅ 4 spaces
```

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10. Vocabulary

English	Vietnamese	Meaning
condition	điều kiện	An expression that is True or False
boolean	kiểu logic	A value that is either True or False
True	đúng	The boolean value for "yes"
False	sai	The boolean value for "no"
comparison	so sánh	Checking if values are equal, greater, etc.
indentation	thụt lề	Spaces at the start of a line to show a block
block	khối lệnh	A group of lines that belong together
branch	nhánh	One path of an if/else decision

English	Vietnamese	Meaning
equal	bằng	Having the same value (`==`)
not equal	không bằng	Having different values (`!=`)
greater than	lớn hơn	One value is bigger (`>`)
less than	nhỏ hơn	One value is smaller (`<`)

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11. Summary

✅ Comparison operators (`==` , `!=` , `<` , `>` , `<=` , `>=`) return `True` or `False` . ✅ `if condition:` runs the indented block only when the condition is `True` . ✅ `else:` runs when the condition is `False` . ✅ **Indentation (4 spaces) is required** — Python uses it to define blocks. ✅ `=` assigns a value. `==` compares two values. ✅ You can use `if` without `else` if you only need to act on the True case.

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12. Homework

🟢 Easy

Exercise E1. Positive or Negative

Read a number. Print `positive` if it is greater than 0, otherwise print `not positive` .

- **Input:** One integer.
- **Output:** `positive` or `not positive` .

Example:

Input	Output
5	positive
-3	not positive
0	not positive

Exercise E2. Even or Odd

Read a whole number. Print `even` if it is divisible by 2, otherwise print `odd` .

- **Input:** One integer.
- **Output:** `even` or `odd`.

Example:

Input	Output
4	even
7	odd
0	even

Exercise E3. Pass or Fail

Read a score (0–100). Print `pass` if the score is 50 or above, otherwise print `fail`.

- **Input:** One integer (0 ≤ score ≤ 100).
- **Output:** `pass` or `fail`.

Example:

Input	Output
75	pass
49	fail
50	pass

Exercise E4. Zero Check

Read a number. Print `zero` if it equals 0, otherwise print `not zero`.

- **Input:** One integer.
- **Output:** `zero` or `not zero`.

Example:

Input	Output
0	zero
5	not zero
-1	not zero

Exercise E5. Adult or Child

Read an age. Print `adult` if the age is 18 or more, otherwise print `child`.

- **Input:** One integer.
- **Output:** `adult` or `child`.

Example:

Input	Output
20	adult
10	child
18	adult

Exercise E6. Bigger Number

Read two numbers. Print `first` if the first number is greater than the second, otherwise print `second`.

- **Input:** Two integers, one per line.
- **Output:** `first` or `second`.

Example:

Input	Output
8 / 3	first
2 / 9	second
5 / 5	second

Exercise E7. Divisible by 5

Read a number. Print `yes` if it is divisible by 5, otherwise print `no`.

- **Input:** One integer.
- **Output:** `yes` or `no`.

Example:

Input	Output
10	yes
7	no
25	yes

🟡 Medium

Exercise M1. Maximum of Two

Read two numbers. Print the larger one. If they are equal, print either one.

- **Input:** Two integers, one per line.
- **Output:** The larger number.

Example:

Input	Output
7 / 3	7
4 / 9	9
5 / 5	5

Exercise M2. Minimum of Two

Read two numbers. Print the smaller one.

- **Input:** Two integers, one per line.
- **Output:** The smaller number.

Example:

Input	Output
7 / 3	3
4 / 9	4

Exercise M3. Temperature Check

Read a temperature in Celsius. Print `hot` if it is above 30, otherwise print `cool`.

- **Input:** One integer.
- **Output:** `hot` or `cool`.

Example:

Input	Output
35	hot
20	cool
30	cool

Exercise M4. Divisible by 3

Read a number. Print `divisible` if it is divisible by 3, otherwise print `not divisible`.

- **Input:** One integer.
- **Output:** `divisible` or `not divisible`.

Example:

Input	Output
9	divisible
10	not divisible
0	divisible

Exercise M5. Same or Different

Read two numbers. Print `same` if they are equal, otherwise print `different`.

- **Input:** Two integers, one per line.
- **Output:** `same` or `different`.

Example:

Input	Output
5 / 5	same
3 / 7	different

Exercise M6. Long or Short

Read a word (string). Print `long` if it has more than 5 characters, otherwise print `short`.

- **Input:** One word (no spaces).
- **Output:** `long` or `short`.

Example:

Input	Output
python	long
cat	short
hello	short

(Hint: Use `len(word)` to get the number of characters.)

Exercise M7. Profit or Loss

Read the buying price and selling price of an item. Print `profit` if the selling price is higher, `loss` if lower, or `break even` if they are equal.

- **Input:** Two integers — buying price, then selling price.
- **Output:** `profit`, `loss`, or `break even`.

Example:

Input	Output
100 / 150	profit
200 / 180	loss
100 / 100	break even

🔴 Hard

Exercise H1. Ticket Price

A cinema has two ticket prices: children under 12 years old pay **50,000 dong**, and everyone else pays **80,000 dong**. Write a program that reads a person's age and prints the ticket price

they must pay.

- **Input:** One integer — the person's age (1 ≤ age ≤ 100).
- **Output:** One integer — the ticket price in dong.

Example:

Input	Output
8	50000
15	80000
12	80000

Explanation: Age 8 < 12, so the child pays 50,000. Age 15 ≥ 12, so the person pays 80,000. Age 12 is not less than 12, so they pay 80,000.

Exercise H2. Absolute Value

In mathematics, the absolute value of a number is its distance from zero — always positive. For example, |–5| = 5 and |3| = 3.

Write a program that reads a number and prints its absolute value **without using the built-in `abs()` function**.

- **Input:** One integer (–1000 ≤ n ≤ 1000).
- **Output:** The absolute value.

Example:

Input	Output
-5	5
3	3
0	0

Explanation: If the number is negative, multiply by –1 to make it positive. If it is already 0 or positive, print it as is.

Exercise H3. Largest of Two

Read two numbers. Print the larger one. If they are equal, print that number.

- **Input:** Two integers, one per line.
- **Output:** The larger number.

Example:

Input	Output
10 / 7	10
3 / 8	8
6 / 6	6

Explanation: Compare the two numbers. The larger one wins. If equal, either one is fine.

Exercise H4. Leap Year

A year is a **leap year** if it is divisible by 4 but NOT divisible by 100, OR if it is divisible by 400. Leap years have 366 days instead of 365.

Write a program that reads a year and prints `leap year` or `not leap year`.

- **Input:** One integer — the year (1 ≤ year ≤ 9999).
- **Output:** `leap year` or `not leap year`.

Example:

Input	Output
2024	leap year
2023	not leap year
1900	not leap year
2000	leap year

Explanation: 2024 ÷ 4 = 506 (divisible by 4) and 2024 ÷ 100 = 20.24 (not divisible by 100) → leap year. 1900 ÷ 100 = 19 (divisible by 100) but 1900 ÷ 400 = 4.75 (not divisible by 400) → not a leap year.

(Hint: Use `if/else` with `%`. You may need to think carefully about the logic. You can use `and` / `or` — we will study them properly in Lesson 7, but you can try them here!)

Exercise H5. Triangle Valid

Three sticks have lengths A, B, and C. They can form a triangle only if the sum of any two sides is **greater than** the third side. This must be true for all three combinations.

Write a program that reads three side lengths and prints `valid` or `invalid`.

- **Input:** Three integers on separate lines — A, B, C (1 ≤ A, B, C ≤ 1000).
- **Output:** `valid` or `invalid`.

Example:

Input	Output
3 / 4 / 5	valid
1 / 2 / 10	invalid
5 / 5 / 5	valid

Explanation: For 3, 4, 5: 3+4=7>5 ✅, 3+5=8>4 ✅, 4+5=9>3 ✅ → valid. For 1, 2, 10: 1+2=3 which is NOT > 10 → invalid.

Exercise H6. Grade

A teacher uses this grading scale: A for 90 and above, B for 80–89, C for 70–79, D for 60–69, F for below 60.

Write a program that reads a score and prints the grade. Use only `if` and `else` (no `elif` — that is Lesson 6!).

- **Input:** One integer — the score (0 ≤ score ≤ 100).
- **Output:** One letter — the grade.

Example:

Input	Output
95	A
83	B
72	C
65	D
45	F

(Hint: Nest `if/else` inside another `else`. It gets a bit long, but it works!)

Exercise H7. Discount

A shop gives a **10% discount** if a customer buys **10 or more** items. Otherwise, no discount. Write a program that reads the price per item and the quantity, then prints the final total price.

- **Input:** Two lines — price per item (integer), then quantity (integer).
- **Output:** The final total price (as a float or integer).

Example:

Input	Output
20000 / 10	180000.0
20000 / 5	100000
15000 / 12	162000.0

Explanation: For 10 items at 20,000 each: total = 200,000. With 10% discount: 200,000 × 0.9 = 180,000.